

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second semester of B.Pharm. Examination October 2017
MA122 Advanced Mathematics

Date: 09.10.2017, Monday

Time: 10:00a.m. to 01:00p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section I are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.
6. Use of scientific calculator is allowed.

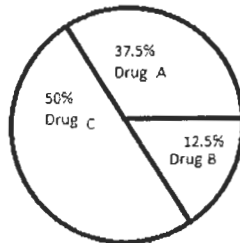
SECTION - I**Q 1** Choose the correct answer for the following questions.**20**

1. Fourth derivative of $y = e^{3x}$ is
[A] $y_4 = 2^4 e^{3x}$.
[B] $y_4 = 4^4 e^{3x}$.
[C] $y_4 = e^{3x}$.
[D] $y_4 = 3^4 e^{3x}$.
2. Second derivative of $y = x^2 + 2x + 3$ is
[A] 1.
[B] 3.
[C] 0.
[D] 2.
3. Which of the following is true?
[A] Every Taylor series is Macluarin series.
[B] Macluarin series expansion around 0 is Taylor series.
[C] Macluarin series expansion arround 1 is Taylor series.
[D] Macluarin series expansion arround 2 is Taylor series.
4. The order and degree of an ordinary differential equation $\frac{d^2y}{dx^2} + 4xy = 3$ are
[A] order 1 and degree 1.
[B] order 2 and degree 1.
[C] order 1 and degree 2.
[D] order2 and degree 2.

5. The first order ordinary differential equation $M(x, y)dx + N(x, y)dy = 0$ is exact if and only if
- [A] $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$.
- [B] $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$.
- [C] $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
- [D] $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$.
6. The general form of linear differential equation is
- [A] $\frac{dy}{dx} + p(y)y = q(x, y)$.
- [B] $\frac{dy}{dx} + p(x, y)y = q(y)$.
- [C] $\frac{dy}{dx} + p(x)y^n = q(x), n > 1$.
- [D] $\frac{dy}{dx} + p(x)y = q(x)$.
7. Order of homogeneous function $f(x, y) = x^2 + y^2$ is
- [A] 1.
- [B] 2.
- [C] 3.
- [D] 4.
8. Which of the following is a particular integral of $y'' + 2y' + y = 0$?
- [A] 0
- [B] 1
- [C] e^x
- [D] e^{-x}
9. If roots of auxiliary equation are $0 + i$ and $0 - i$, then corresponding component (basis) solutions of a 2nd order ordinary differential equation are
- [A] e^x & e^{2x} .
- [B] e^{-x} & e^x .
- [C] $\cos x$ & $\sin x$.
- [D] $e^x \cos x$ & $e^x \sin x$.
10. Which of the following is true for the rate of change in number of bacteria(x) with respect to time(t) is inverse propositional to t^2 ?
- [A] $t \frac{dx}{dt} = m$
- [B] $t^2 \frac{dx}{dt} = m$
- [C] $\frac{dx}{dt} = mt^2$
- [D] $\frac{dx}{dt} = mt + 1$

11. The rate at which bacteria multiply is proportional to the current number of bacteria. if x be the number of bacteria present at time t , then
- [A] $\frac{dx}{dt} = mx$.
- [B] $\frac{dx}{dt} = mx^2$.
- [C] $\frac{dx}{dt} = mx + kt$.
- [D] $\frac{dx}{dt} = e^{mx}$.
12. If T be the temperature of the body at time t and t_s is the temperature of surrounding, then Newton's law of cooling is given by
- [A] $\frac{dT}{dt} = k(T - t_s)$.
- [B] $\frac{dT}{dt} = kT + t_s$.
- [C] $\frac{dT}{dt} = kT$.
- [D] $\frac{dT}{dt} = k(t_s - T)$.
13. Equation of a straight line passing through $(1, 1)$ and $(3, -1)$ is
- [A] $x + y = 2$.
- [B] $x - y = 2$.
- [C] $3x + y = 0$.
- [D] $x - 3y = 0$.
14. Equation of a straight line parallel to $y = 5x - 8$ is
- [A] $5y + 5x = 3$.
- [B] $y = 5x + 8$.
- [C] $y = 2x + 3$.
- [D] $y = 4x + 5$.
15. Two lines with slope m_1 & m_2 are said to be perpendicular if
- [A] $m_1 m_2 = 1$.
- [B] $m_1 - m_2 = 0$.
- [C] $m_1 + m_2 = 1$.
- [D] $m_1 m_2 = -1$.
16. Distance between two points $(3, 4)$ and $(3, 5)$ is
- [A] 0.
- [B] 1.
- [C] -1.
- [D] 3.
17. Mean of data set 2, 5, 5, 9, 10, 11, 12, 12 and 15 is
- [A] 7.
- [B] 8.
- [C] 9.
- [D] 10.

18. Data set 1, 5, 7, a , 12, 14 and 17 is arranged in increasing order. If Median of a following data set is 9, then a is
 [A] 8.
 [B] 9.
 [C] 10.
 [D] 11.
19. Following graph shows production of different types of drug by a company in a month. If total production of drugs by a company is 24 kg in one month, then production of drug C in a month is



- [A] 6 kg.
 [B] 9 kg.
 [C] 12 kg.
 [D] 15 kg.
20. If mean of data set 5, 8, 10, a , 16, 18, 20 is 13, then value of a is
 [A] 10.
 [B] 12.
 [C] 14.
 [D] 15.

SECTION – II

Q 2 Attempt any **TWO** of the following

- A** If rate of a change of number of Bacteria(x) with respect to t -time(hour) is 05
 given by $\frac{dx}{dt} = 3xt^2$, then find number of Bacteria at $t = 2$ hours and at initial stage $t = 0$ and $x = 0$.
- B** Find mean, median and relative frequency distribution(%) of ungrouped data 05
 set 1, 4, 5, 3, 2, 5, 8, 1 and 2.
- C** Derive equation of the distance between two points and using it find distance 05
 between (8, -1) and (2, 3).

Q 3 Attempt any **TWO** of the following

- A** Verify Cauchy's mean value theorem for $f(x) = \log x$ and $g(x) = \frac{1}{x}$, $x \in [1, 2]$. 05
- B** Identify the type of an ordinary differential equation $\frac{dy}{dx} + \frac{y}{x} = x$ and find 05
 particular solution with help of initial condition $y(0) = 0$.
- C** Find general solution of $y'' + 6y' + 9y = 0$. 05

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Bacteria multiply at the rate proportional to the number present. If the original number N doubles in three hours, find in how many hours, the numbers of the bacteria will be $4N$. **05**
- B** If the temperature of a body drops from 100° to 60° in one minute when the temperature of the surrounding is 20° , what will be the temperature of the body at the end of the second minute? **05**
- C** Find Mean for following ungrouped data 2, 5, 12, 10, 20, 15, 18, 26, 25, 36, 34, 30 and 42. Also organize the above data into groups with 5 intervals starting from smallest value and using it evaluate relative frequency(%) of each class. **05**
- D** A rectangle having vertices $A(1, 1)$, $B(1, 2)$, $C(2, 1)$ and $D(2, 2)$ is given. Find length, height and area of rectangle. **05**
- E** If rate of change of population is proportional to time and at 1 hour rate is given by 150 person per hour, then find number of person at $t = 2$ hour where at initial stage number of person is zero. **05**
- F** (I) State the governing differential equation of Newton's law of cooling and solve it. **03**
- (II) Describe mean, median and mode. **02**
- Q 5** Attempt any **FOUR** of the following
- A** State Taylor's theorem for expansion of a function and expand $f(x) = \sin x$ at $x = \pi/2$. **05**
- B** State Maclaurin's Series and find Maclaurin's Series of $f(x) = \log(1+x)$. **05**
- C** Solve: $\frac{dy}{dx} - 2\frac{y}{x} = x^2$ **05**
- D** Show that $2xydx + (x^2 + y^2 + 3)dy = 0$ is an exact differential equation and obtain general solution of it. **05**
- E** Find particular integral of a differential equation having basis solutions $y_1 = e^{2x}$, $y_2 = e^{3x}$ and $R(x) = e^{3x}$. **05**
- F** Solve: $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 2y = 0$. **05**
